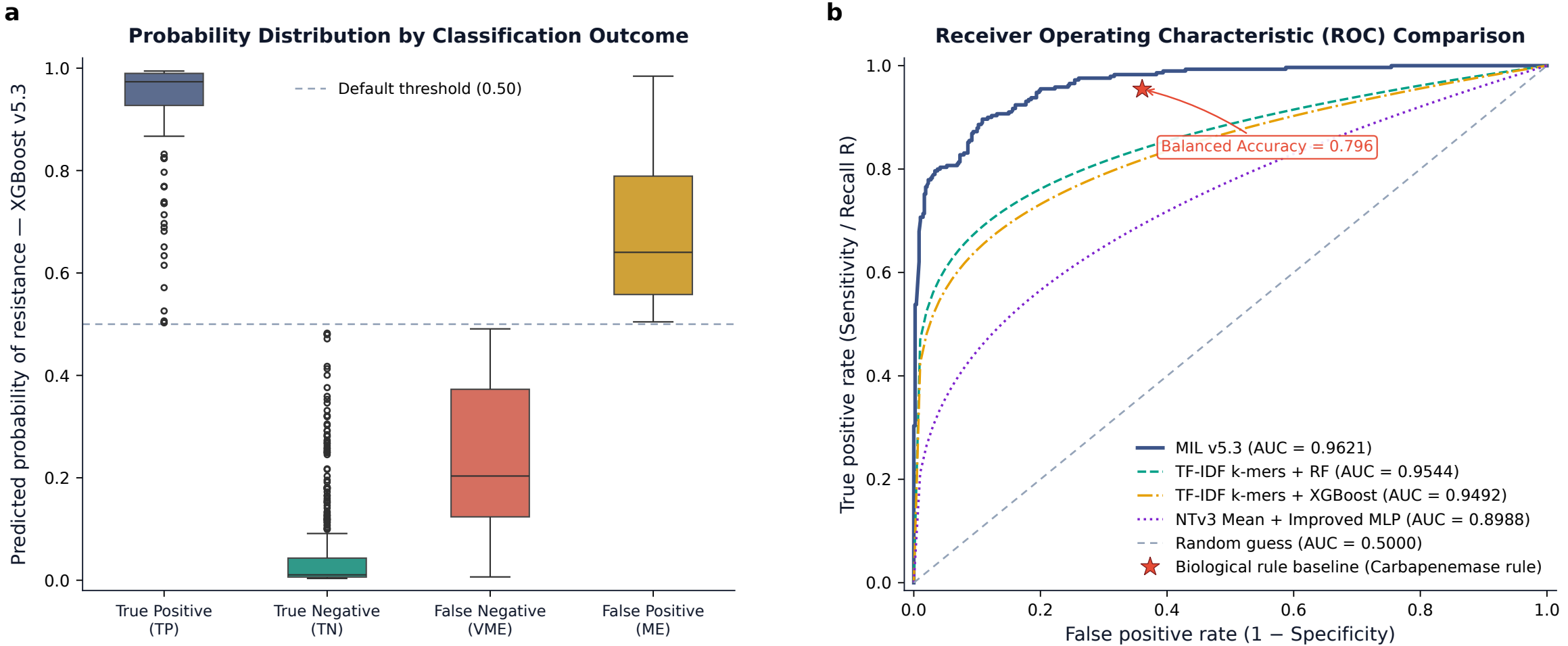
a
Probability Distribution by Classification Outcome
Predicted probability of resistance — XGBoost v5.3
Default threshold (0.50)
1.0
0.8
0.6
0.4
0.2
0.0
True Positive (TP)
True Negative (TN)
False Negative (VME)
False Positive (ME)
b
Receiver Operating Characteristic (ROC) Comparison
True positive rate (Sensitivity / Recall R)
False positive rate (1 − Specificity)
Balanced Accuracy = 0.796
MIL v5.3 (AUC = 0.9621)
TF-IDF k-mers + RF (AUC = 0.9544)
TF-IDF k-mers + XGBoost (AUC = 0.9492)
NTv3 Mean + Improved MLP (AUC = 0.8988)
Random guess (AUC = 0.5000)
Biological rule baseline (Carbapenemase rule)